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Network video streaming with trick play based on separate trick play files

Abstract

Network services encode multimedia content, such as video, into multiple adaptive bitrate streams of encoded video and a separate trick play stream of encoded video to support trick play features. The trick play stream is encoded at a lower encoding bitrate and frame rate than each of the adaptive bitrate streams. The adaptive bitrate streams and the trick play stream are stored in the network services. During normal content streaming and playback, a client device downloads a selected one of the adaptive bitrate streams from network serviced for playback at the client device. To implement a trick play feature, the client device downloads the trick play stream from the network services for trick play playback.

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7974714	12/2010	Hoffberg	N/A	N/A
7984513	12/2010	Kyne et al.	N/A	N/A
7991156	12/2010	Miller	N/A	N/A
8001471	12/2010	Shaver et al.	N/A	N/A
8010810	12/2010	Fitzgerald et al.	N/A	N/A
8015491	12/2010	Shaver et al.	N/A	N/A
8023562	12/2010	Zheludkov et al.	N/A	N/A
8046453	12/2010	Olaiya	N/A	N/A
8054880	12/2010	Yu et al.	N/A	N/A
8065708	12/2010	Smyth et al.	N/A	N/A
8069260	12/2010	Speicher et al.	N/A	N/A
8073900	12/2010	Guedalia et al.	N/A	N/A
8074083	12/2010	Lee et al.	N/A	N/A
8078644	12/2010	Hannuksela	N/A	N/A
8082442	12/2010	Keljo et al.	N/A	N/A
8131875	12/2011	Chen	N/A	N/A
8135041	12/2011	Ramaswamy	N/A	N/A
8160157	12/2011	Lamy-Bergot et al.	N/A	N/A
8169916	12/2011	Pai et al.	N/A	N/A
8170210	12/2011	Manders et al.	N/A	N/A
8195714	12/2011	Mckibben et al.	N/A	N/A
8201264	12/2011	Grab et al.	N/A	N/A
8213607	12/2011	Rose et al.	N/A	N/A
8213768	12/2011	Morioka et al.	N/A	N/A
8218439	12/2011	Deshpande	N/A	N/A
8225061	12/2011	Greenebaum	N/A	N/A
8233768	12/2011	Soroushian et al.	N/A	N/A
8243924	12/2011	Chen et al.	N/A	N/A
8245124	12/2011	Gupta	N/A	N/A
8249168	12/2011	Graves	N/A	N/A
8261356	12/2011	Choi et al.	N/A	N/A
8265168	12/2011	Masterson et al.	N/A	N/A
8270473	12/2011	Chen et al.	N/A	N/A
8270819	12/2011	Vannier	N/A	N/A
8275871	12/2011	Ram et al.	N/A	N/A
8286213	12/2011	Seo	N/A	N/A
8286621	12/2011	Halmone	N/A	N/A
8289338	12/2011	Priyadarshi et al.	N/A	N/A
8290157	12/2011	Candelore	N/A	N/A
8291460	12/2011	Peacock	N/A	N/A
8296434	12/2011	Miller et al.	N/A	N/A
8311094	12/2011	Kamariotis et al.	N/A	N/A
8311111	12/2011	Xu et al.	N/A	N/A
8311115	12/2011	Gu et al.	N/A	N/A
8312079	12/2011	Newsome et al.	N/A	N/A
8321556	12/2011	Chatterjee et al.	N/A	N/A
8325800	12/2011	Holcomb et al.	N/A	N/A
8327009	12/2011	Prestenback et al.	N/A	N/A
8341715	12/2011	Sherkin et al.	N/A	N/A
8346753	12/2012	Hayes	N/A	N/A
8365235	12/2012	Hunt et al.	N/A	N/A
8369421	12/2012	Kadono et al.	N/A	N/A
8380041	12/2012	Barton et al.	N/A	N/A
8386621	12/2012	Park	N/A	N/A
8396114	12/2012	Gu et al.	N/A	N/A
8397265	12/2012	Henocq et al.	N/A	N/A
8401188	12/2012	Swaminathan	N/A	N/A
8401900	12/2012	Cansler et al.	N/A	N/A
8407753	12/2012	Kuo	N/A	N/A
8412841	12/2012	Swaminathan et al.	N/A	N/A

8423889	12/2012	Zagorie et al.	N/A	N/A
8452110	12/2012	Carmel et al.	N/A	N/A
8456380	12/2012	Pagan	N/A	N/A
8464066	12/2012	Price et al.	N/A	N/A
8472792	12/2012	Butt	N/A	N/A
8473630	12/2012	Galligan	N/A	N/A
8484368	12/2012	Robert et al.	N/A	N/A
8510303	12/2012	Soroushian et al.	N/A	N/A
8510404	12/2012	Carmel et al.	N/A	N/A
8514926	12/2012	Ro et al.	N/A	N/A
8515265	12/2012	Kwon et al.	N/A	N/A
8516529	12/2012	Lajoie et al.	N/A	N/A
8526610	12/2012	Shamoon et al.	N/A	N/A
8527645	12/2012	Proffit et al.	N/A	N/A
8543842	12/2012	Ginter et al.	N/A	N/A
8555329	12/2012	Fröjdjdh et al.	N/A	N/A
8571993	12/2012	Kocher et al.	N/A	N/A
8595378	12/2012	Cohn et al.	N/A	N/A
8606069	12/2012	Okubo et al.	N/A	N/A
8630419	12/2013	Mori	N/A	N/A
8631247	12/2013	O'loughlin et al.	N/A	N/A
8640166	12/2013	Craner et al.	N/A	N/A
8649669	12/2013	Braness et al.	N/A	N/A
8650599	12/2013	Shindo et al.	N/A	N/A
8656183	12/2013	Russell et al.	N/A	N/A
8677428	12/2013	Lewis et al.	N/A	N/A
8681866	12/2013	Jia	N/A	N/A
8683066	12/2013	Hurst et al.	N/A	N/A
8689267	12/2013	Hunt	N/A	N/A
8726264	12/2013	Allen et al.	N/A	N/A
8731193	12/2013	Farkash et al.	N/A	N/A
8731369	12/2013	Li et al.	N/A	N/A
RE45052	12/2013	Li	N/A	N/A
8767825	12/2013	Wang et al.	N/A	N/A
8774609	12/2013	Drake et al.	N/A	N/A
8781122	12/2013	Chan et al.	N/A	N/A
8782268	12/2013	Pyle et al.	N/A	N/A
8804956	12/2013	Hiriart	N/A	N/A
8805109	12/2013	Shoham et al.	N/A	N/A
8806188	12/2013	Braness et al.	N/A	N/A
8818171	12/2013	Soroushian et al.	N/A	N/A
8818896	12/2013	Candelore	N/A	N/A
8819116	12/2013	Tomay et al.	N/A	N/A
8832434	12/2013	Apostolopoulos et al.	N/A	N/A
8843586	12/2013	Pantos et al.	N/A	N/A
8849950	12/2013	Stockhammer et al.	N/A	N/A
8850205	12/2013	Choi et al.	N/A	N/A
8850498	12/2013	Roach et al.	N/A	N/A
8856218	12/2013	Inskip	N/A	N/A
8897370	12/2013	Wang et al.	N/A	N/A
8908984	12/2013	Shoham et al.	N/A	N/A
8909922	12/2013	Kiefer et al.	N/A	N/A
8914534	12/2013	Braness et al.	N/A	N/A
8914836	12/2013	Shivadas et al.	N/A	N/A
8918533	12/2013	Chen et al.	N/A	N/A
8918535	12/2013	Ma et al.	N/A	N/A
8918636	12/2013	Kiefer	N/A	N/A
8918908	12/2013	Ziskind et al.	N/A	N/A
8948249	12/2014	Sun et al.	N/A	N/A
8964977	12/2014	Ziskind et al.	N/A	N/A
8997161	12/2014	Priyadarshi et al.	N/A	N/A
8997254	12/2014	Amidei et al.	N/A	N/A
9014471	12/2014	Shoham et al.	N/A	N/A
9015782	12/2014	Acharya et al.	N/A	N/A
9025659	12/2014	Soroushian et al.	N/A	N/A
9038116	12/2014	Knox et al.	N/A	N/A
9038121	12/2014	Kienzle et al.	N/A	N/A
9042670	12/2014	Carmel et al.	N/A	N/A
9049497	12/2014	Chen et al.	N/A	N/A
9060207	12/2014	Scherkus et al.	N/A	N/A
9094737	12/2014	Shivadas et al.	N/A	N/A
9098335	12/2014	Muthiah et al.	N/A	N/A
9111098	12/2014	Smith et al.	N/A	N/A
9124773	12/2014	Chan et al.	N/A	N/A
9125073	12/2014	Oyman et al.	N/A	N/A

9184920	12/2014	Grab et al.	N/A	N/A
9191151	12/2014	Luby et al.	N/A	N/A
9191457	12/2014	Van der Schaar	N/A	N/A
9197685	12/2014	Soroushian	N/A	N/A
9201922	12/2014	Soroushian et al.	N/A	N/A
9203816	12/2014	Brueck et al.	N/A	N/A
9210481	12/2014	Braness et al.	N/A	N/A
9215466	12/2014	Zhai et al.	N/A	N/A
9247311	12/2015	Kiefer	N/A	N/A
9247312	12/2015	Braness et al.	N/A	N/A
9247317	12/2015	Shivadas et al.	N/A	N/A
9253178	12/2015	Blom et al.	N/A	N/A
9264475	12/2015	Shivadas et al.	N/A	N/A
9294531	12/2015	Zhang et al.	N/A	N/A
9313510	12/2015	Shivadas et al.	N/A	N/A
9343112	12/2015	Amidei et al.	N/A	N/A
9344517	12/2015	Shivadas et al.	N/A	N/A
9344721	12/2015	Dikvall	N/A	N/A
9380096	12/2015	Luby et al.	N/A	N/A
9386064	12/2015	Luby et al.	N/A	N/A
9406066	12/2015	Candalore	N/A	N/A
9467708	12/2015	Soroushian et al.	N/A	N/A
9479805	12/2015	Rothschild et al.	N/A	N/A
9485469	12/2015	Kahn et al.	N/A	N/A
9485546	12/2015	Chen et al.	N/A	N/A
9510031	12/2015	Soroushian et al.	N/A	N/A
9571827	12/2016	Su et al.	N/A	N/A
9584557	12/2016	Panje et al.	N/A	N/A
9584847	12/2016	Ma et al.	N/A	N/A
9615061	12/2016	Carney et al.	N/A	N/A
9621522	12/2016	Kiefer et al.	N/A	N/A
9628536	12/2016	Luby et al.	N/A	N/A
9667684	12/2016	Ziskind et al.	N/A	N/A
9672286	12/2016	Soroushian et al.	N/A	N/A
9674254	12/2016	Pare et al.	N/A	N/A
9686332	12/2016	Binns et al.	N/A	N/A
9706259	12/2016	Chan et al.	N/A	N/A
9712890	12/2016	Shivadas et al.	N/A	N/A
9761274	12/2016	Delpuch et al.	N/A	N/A
9798863	12/2016	Grab et al.	N/A	N/A
9813740	12/2016	Panje et al.	N/A	N/A
9866878	12/2017	van der Schaar et al.	N/A	N/A
9883204	12/2017	Braness et al.	N/A	N/A
9906785	12/2017	Naletov et al.	N/A	N/A
9955195	12/2017	Soroushian	N/A	N/A
9967189	12/2017	Patel et al.	N/A	N/A
9967305	12/2017	Braness	N/A	N/A
9967521	12/2017	Kahn et al.	N/A	N/A
10169094	12/2018	Iyer	N/A	N/A
10171873	12/2018	Krebs	N/A	N/A
10212486	12/2018	Chan et al.	N/A	N/A
10225299	12/2018	van der Schaar et al.	N/A	N/A
10225588	12/2018	Kiefer et al.	N/A	N/A
10244272	12/2018	Kiefer et al.	N/A	N/A
10264255	12/2018	Naletov et al.	N/A	N/A
10321168	12/2018	van der Schaar et al.	N/A	N/A
10341698	12/2018	Kiefer et al.	N/A	N/A
10368096	12/2018	Braness et al.	N/A	N/A
10382785	12/2018	Braness et al.	N/A	N/A
10437896	12/2018	Soroushian et al.	N/A	N/A
10462537	12/2018	Shivadas et al.	N/A	N/A
10484749	12/2018	Chan et al.	N/A	N/A
10645429	12/2019	Soroushian	N/A	N/A
10708587	12/2019	Soroushian et al.	N/A	N/A
10715806	12/2019	Naletov et al.	N/A	N/A
10798143	12/2019	Soroushian et al.	N/A	N/A
10805368	12/2019	Van der Schaar et al.	N/A	N/A
10856020	12/2019	Kiefer et al.	N/A	N/A
10878065	12/2019	Grab et al.	N/A	N/A
10893305	12/2020	Van der Schaar et al.	N/A	N/A
10931982	12/2020	Soroushian	N/A	N/A
10992955	12/2020	Braness et al.	N/A	N/A
11102553	12/2020	Chan et al.	N/A	N/A
RE48761	12/2020	Shivadas et al.	N/A	N/A
11438394	12/2021	Van Der Schaar et al.	N/A	N/A

11457054	12/2021	Soroushian et al.	N/A	N/A
11470405	12/2021	Shivadas et al.	N/A	N/A
11638033	12/2022	Braness et al.	N/A	N/A
11683542	12/2022	Kiefer et al.	N/A	N/A
2001/0021276	12/2000	Zhou	N/A	N/A
2001/0030710	12/2000	Werner	N/A	N/A
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2002/0147980	12/2001	Satoda	N/A	N/A
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2002/0161462	12/2001	Fay	N/A	N/A
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2004/0052501	12/2003	Tam	N/A	N/A
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2004/0117347	12/2003	Seo et al.	N/A	N/A
2004/0136698	12/2003	Mock	N/A	N/A
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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The current application is a continuation of U.S. patent application Ser. No. 16/665,652 entitled “Network Video Streaming with Trick Play Based on Separate Trick Play Files” to Shivadas et al., filed Oct. 28, 2019, which is a

continuation of U.S. patent application Ser. No. 15/651,817 entitled “Network Video Streaming with Trick Play Based on Separate Trick Play Files” to Shivadas et al., filed Jul. 17, 2017 and issued as U.S. Pat. No. 10,462,537 on Oct. 29, 2019, which is a continuation of U.S. patent application Ser. No. 14/810,345 entitled “Network Video Streaming with Trick Play Based on Separate Trick Play Files” to Shivadas et al., filed Jul. 27, 2015 and issued as U.S. Pat. No. 9,712,890 on Jul. 18, 2017, which is a continuation of U.S. patent application Ser. No. 13/905,852 entitled “Network Video Streaming with Trick Play Based on Separate Trick Play Files” to Shivadas et al., filed May 30, 2013 and issued as U.S. Pat. No. 9,094,737 on Jul. 28, 2015, the disclosures of which are hereby incorporated by reference in their entireties.

BACKGROUND

(1) Distribution of multimedia video (also referred to herein as “media” and/or “program(s)”), such as movies and the like, from network services to a client device, may be achieved through adaptive bitrate streaming of the video. Prior to streaming, the video may be encoded at different bitrates and resolutions into multiple bitrate streams that are stored in the network services. Typically, each of the bitstreams includes time-ordered segments of encoded video.

(2) Adaptive bitrate streaming includes determining an available streaming bandwidth at the client device, and then downloading a selected one of the different bitrate streams from the network services to the client device based on the determined available bandwidth. While streaming, the client device downloads and buffers the successive encoded video segments associated with the selected bitstream. The client device decodes the buffered encoded video segments to recover the video therein, and then plays back the recovered video on the client device, e.g., in audio-visual form.

(3) In normal playback, the client device plays back the video recovered from each of the buffered segments in the order in which the video was originally encoded, i.e., in a forward direction. The client device may offer playback modes or features in addition to normal playback. Such additional playback features may include rewind, fast forward, skip, and so on, as is known.

(4) The additional playback features are referred to herein as trick play features. In order to implement trick play features, such as rewind, the client device requires access to video that has already been played. Therefore, the client device may be required to store large amounts of already downloaded and played video in order to meet the demands of a selected trick play feature. However, many client devices, especially small, hand-held devices, have limited memory capacity and, therefore, may be unable to store the requisite amount of video.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagram of an example network environment that supports adaptive bitrate streaming of multimedia content, such as video, with trick play features.

(2) FIG. 2 is an illustration of an example encoded multimedia video program generated by and stored in network services of FIG. 1.

(3) FIG. 3A is an illustration of an example adaptive bitrate frame structure of an encoded video block of FIG. 2.

(4) FIG. 3B is an illustration of an example trick play frame structure of an encoded video block of FIG. 2.

(5) FIG. 4 is a sequence diagram of example high-level interactions between network services and a client device used to initiate streaming, implement normal streaming and playback, and implement trick play features in streaming embodiments.

(6) FIG. 5 is an example Profile message used in streaming.

(7) FIG. 6 is an example Playlist message used in streaming.

(8) FIG. 7 is a flowchart of an example network-side method of multimedia content streaming with trick play support based on trick play files, which may be implemented in the network services of FIG. 1.

(9) FIG. 8 is a flowchart of an example client-side method of multimedia content streaming with trick play support based on trick play files, which may be implemented in the client device of FIG. 1.

(10) FIG. 9A is a block diagram of an example computer system.

(11) FIG. 9B is a block diagram of network/server-side application instructions which may execute in on a processor system similar to that of FIG. 9A.

(12) FIG. 10 is a block diagram of an example computer system corresponding to any of the network servers in the environment of FIG. 1.

(13) FIG. 11 is a block diagram of an example system representing a client device of FIG. 1.

(14) In the drawings, the leftmost digit(s) of a reference number identifies the drawing in which the reference number first appears.

DETAILED DESCRIPTION

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1 Network Environment

(16) FIG. 1 is a block diagram of an example network environment **100** that supports adaptive bitrate streaming of multimedia content with trick play features. Network services **102** encode multimedia content, such as video, into multiple adaptive bitrate streams of encoded video and a separate trick play stream of encoded video to support trick play features. The trick play stream may be encoded at a lower encoding bitrate and a lower frame than each of the adaptive bitrate streams. The adaptive bitrate and trick play streams are stored in network services **102**. For normal content streaming and playback, a client device **104** downloads a selected one of the adaptive bitrate streams from network services **102** for playback at the client device. When a user of client device **104** selects a trick play feature, such as rewind, the client device **104** downloads the trick play stream from network services **102** for trick play playback.

(17) Environment **100** supports trick play features in different adaptive bitrate streaming embodiments, including on-demand streaming, live streaming, and real-time streaming embodiments. On-demand streaming includes encoding the content of a program from start to end in its entirety and then, after the entire program has been encoded, streaming, i.e., downloading, the encoded program to a client device. An example of on-demand streaming includes streaming a movie from a Video-on-Demand (VOD) service to a client device.

(18) Live streaming includes encoding successive blocks of live content, i.e., a live program, as they are received from a content source, and then streaming each encoded block as it becomes available for download. Live streaming may include streaming live scenes, i.e., video, captured with a video camera.

(19) Real-time streaming is similar in most aspects to live streaming, except that the input to real-time streaming is not a live video feed. Rather, the input, or source, may include successive encoded blocks, or input blocks, that have a format not suitable for streaming (e.g., for a given system) and must, therefore, be decoded and re-encoded (i.e., transcoded) into an encoded format that is suitable for streaming (in the given system). Real-time streaming handles the successive incompatible input blocks similar to the way live streaming handles the successive blocks of live content.

(20) Network environment **100** is now described in detail. Network environment **100** includes server-side or network services **102** (also referred to simply as “services **102**”) and client-side device **104**. Network services **102** may be implemented as Internet cloud-based services. Network services **102** interact and cooperate with each other, and with client device **104**, to manage and distribute, e.g., stream, multimedia content from content sources **108** to the client devices, over one or more communication network **106**, such as the Internet. Network services **102** communicate with each other and with client devices **104** using any suitable communication protocol, such as an Internet protocol, which may include Transmission Control

Protocol/Internet Protocol (TCP/IP), Hypertext Transfer Protocol (HTTP), etc., and other non-limiting protocols described herein.

(21) Content sources **108** may include any number of multimedia content sources or providers that originate live and/or pre-recorded multimedia content (also referred to herein simply as “content”), and provide the content to services **102**, directly, or indirectly through communication network **106**. Content sources **108**, such as Netflix®, HBO®, cable and television networks, and so on, may provide their content in the form of programs, including, but not limited to, entertainment programs (e.g., television shows, movies, cartoons, news programs, etc.), educational programs (e.g., classroom video, adult education video, learning programs, etc.), and advertising programs (e.g., commercials, infomercials, or marketing content). Content sources **108**, such as, e.g., video cameras, may capture live scenes provide the resulting real-time video to services **102**. Content sources may also include live broadcast feeds deployed using protocols such as Real-time Transport Protocol (RTP), and Real-time Messaging Protocol (RTMP). (22) Network services **102** include, but are not limited to: an encoder **110** to encode content from content sources **108**; a content delivery network (CDN) **112** (also referred to as a “download server **112**”) to store the encoded content, and from which the stored, encoded content may be streamed or downloaded to client device **104**; and a real-time service (RTS) **114** (also referred to as a “real-time server (RTS) **114**”) to (i) control services **102**, and (ii) implement an RTS streaming control interface through which client device **104** may initiate and then monitor both on-demand, live, and real-time streaming sessions. Each of services **102** may be implemented as one or more distinct computer servers that execute one or more associated server-side computer program applications suited to the given service.

(23) Encoder **110** may be implemented as a cloud encoder accessible over communication network **106**. Encoder **110** encodes content provided thereto into a number of alternative bitstreams **120** (also referred to as encoded content) to support adaptive bitrate streaming of the content. For increased efficiency, encoder **110** may be implemented as a parallel encoder that includes multiple parallel encoders. In such an embodiment, encoder **110** divides the content into successive blocks or clips each of a limited duration in time. Each block may include a number of successive picture frames, referred to collectively as a group of pictures (GOPs). Encoder **110** encodes the divided blocks or GOPs in parallel to produce alternative bitstreams **120**. Encoder **110** may also include transcoders to transcode input files from one encoded format to another, as necessary.

(24) Alternative bitstreams **120** encode the same content in accordance with different encoding parameters/settings, such as at different encoding bitrates, resolutions, frame rates, and so on. In an embodiment, each of bitstreams **120** comprises a large number of sequential (i.e., time-ordered) files of encoded content, referred to herein as container files (CFs), as will be described further in connection with FIG. 2.

(25) After encoder **110** has finished encoding content, e.g., after each of the content blocks is encoded, the encoder uploads the encoded content to CDN **112** for storage therein. CDN **112** includes one or more download servers (DSs) to store the uploaded container files at corresponding network addresses, so as to be accessible to client device **104** over communication network **106**.

(26) RTS **114** acts as a contact/control point in network services **102** for client device **104**, through which the client device may initiate and then monitor its respective on-demand, live, and real-time streaming sessions. To this end, RTS **114** collects information from services **102**, e.g., from encoder **110** and CDN **112**, that client device **104** may use to manage its respective streaming sessions, and provides the collected information to the client device via messages (described below) when appropriate during streaming sessions, thus enabling the client device to manage its streaming sessions. The information collected by RTS **114** (and provided to client device **104**) identifies the encoded content, e.g., the container files, stored in CDN **112**, and may include, but is not limited to, network addresses of the container files stored in the CDN, encoding parameters used to encode the container files, such as their encoding bitrates, resolutions, and video frame rates, and file information, such as file sizes, and file types.

(27) Client device **104** may be capable of wireless and/or wired communication with network services **102** over communication network **106**, and includes processing, storage, communication, and user interface capabilities sufficient to provide all of the client device functionality described herein. Such functionality may be provided, at least in part, by one or more client applications **107**, such as computer programs, that execute on client device **104**. Client applications **107** may include: a. a Graphical User Interface (GUI) through which a user of the client device may interact with and request services from corresponding server-side applications hosted in services **102**. The GUI may also present trick play feature selections to the user, such as rewind and fast forward. Under user control through the GUI, client device **104** may request/select (i) programs to be streamed from services **102**, and (ii) trick play features to control trick play playback of the streamed programs; b. streaming and playback applications to stream/download the selected programs from the services, and playback, i.e., present, the streamed programs on client device **104**, under user control, through the GUI; and c. a trick play application, integrated with the GUI and the streaming and playback applications, to implement the trick play features as described herein.

2 Container Files—Streaming Sources

(28) As described above, encoder **110** encodes multimedia content from content sources **108**, and CDN **112** stores the encoded content. To support adaptive bitrate streaming and trick play features, encoder **110** encodes the content at multiple encoding levels, where each level represents a distinct combination of an encoding bitrate, a video resolution (for video content), and a video frame rate, to produce (i) multiple adaptive bitrate streams for the content, and (ii) a trick play stream for the content. The multiple streams may be indexed according to their respective encoding levels. While streaming the encoded program from CDN **112**, client device **104** may switch between streams, i.e., levels (and thus encoded bitrates and resolutions), according to conditions at the client device. Also, while streaming the encoded program, client device **104** may download portions of the trick play stream from CDN **112** to implement trick play features in the client device.

(29) FIG. 2 is an illustration of an example encoded multimedia video program **200** generated by encoder **110** and stored in CDN **112**. Encoded video program **200** includes: a. two encoded adaptive bitrate (ABR) video streams 1, 2 encoded at corresponding encoding levels L1, L2 and available for adaptive bitrate streaming; and b. a trick play stream encoded at an encoding level L3. The trick play stream corresponds to, i.e., encodes the same video as, the two ABR streams 1, 2.

(30) Each of encoding levels L1-L3 corresponds to a distinct combination of an encoding bitrate (Rate), a video resolution (Res), and a video frame rate (FR). In the example, encoding levels L1, L2, L3 correspond to encoder settings Rate1/Res1/FR1, Rate2/Res2/FR2, Rate3/Res3/FR3, respectively. In an embodiment, the encoding bitrate Rate3 and the video frame rate FR3 used to encode the trick play stream are less than the encoding bitrates Rate1, Rate2 and the frame rates FR1, FR2, respectively, used to encode adaptive bitrate streams 1, 2.

(31) Although the example of FIG. 2 includes only two encoding levels for the ABR streams, in practice, an encoded video program typically includes many more than two levels of encoding for ABR streaming, such as 8 to 15 levels of encoding.

(32) Each of streams 1-3 includes a distinct, time-ordered, sequence of container files CF (i.e., successive container files CF), where time is depicted in FIG. 2 as increasing in a downward vertical direction. Each of the successive container files CF, of each of streams 1-3, includes (i.e., encodes) a block or segment of video (also referred to herein as an encoded video block or segment) so that the successive container files encode successive contiguous encoded video blocks. Each of container files CF includes a time code TC to indicate a duration of the video encoded in the block of the container file, and/or a position of the container file in the succession of container files comprising the corresponding stream. The time code TC may include a start time and end time for the corresponding encoded video block. In an example in which each of container files CF encodes two seconds of video, time codes TC1, TC2, and TC3 may represent start and end times of 0 s (seconds) and 2 s, 2 s and 4 s, and 4 s and 6 s, respectively, and so down the chain of remaining successive container files.

(33) The encoded blocks of the container files CF in a given stream may encode the same content (e.g., video content) as corresponding blocks in the other streams. For example, the stream 1 block corresponding to time code TC1 has encoded therein the same video as that in the stream 2 block corresponding to TC1. Such corresponding blocks encode the same content and share the same time code TC, i.e., they are aligned or coincide in time.

(34) In an embodiment, a program stream index **204** may be associated with encoded video program **200** to identify each of the streams therein (e.g., the ABR streams 1, 2, and the trick play stream). RTS **114** may create (and store) program stream index **204** based on the information collected from

encoder **110** and CDN **112** as described above in connection with FIG. 1. Then, during a live streaming session, for example, RTS **114** may provide information from program stream index **204** to client device **104** so as to identify appropriate container file addresses to the client device. Program stream index **204** may include: a. address pointers (e.g., network addresses, such as Uniform Resource Locators (URLs)) **210-1**, **210-2**, **210-3** to corresponding streams 1, 2, and the trick play stream; b. encoder parameters/settings associated with the encoded streams including, but not limited to, encoding levels L1, L2, L3 (also referred to as "Video ID" in FIG. 2, and including the encoding bitrates and resolutions Rate1/Res1, Rate2/Res2, Rate3/Res3), encoding techniques/standards, and file types and sizes of the container files CF; and c. a trick play flag (TP flag) associated with URL **210-3** that, when set, indicates the associated stream is a trick play stream.

(35) Address pointers **210-1**, **210-2**, **210-3** may point to respective lists of addresses A1, A2, A3 of the container files CF comprising each of streams 1, 2, 3. Address lists A1, A2, A3 may each be represented as an array or linked list of container file network addresses, e.g., URLs. Accordingly, access to the information in program stream index **204** results in possible access to all of the container files associated with streams 1, 2, 3.

(36) Although each of container files CF depicted in FIG. 2 represents a relatively small and simple container structure, larger and more complicated container structures are possible. For example, each container file may be expanded to include multiple clusters of encoded media, each cluster including multiple blocks of encoded media, to thereby form a larger container file also suitable for embodiments described herein. The larger container files encode an equivalent amount of content as a collection of many smaller container files.

(37) Container files may encode a single stream, such as a video stream (as depicted in FIG. 2), an audio stream, or a text stream (e.g., subtitles). Alternatively, each container file may encode multiple multiplexed streams, such as a mix of video, audio, and text streams. In addition, a container file may encode only a metadata stream at a relatively low bitrate.

(38) In embodiments: the container files may be Matroska (MKV) containers based on Extensible Binary Meta Language (EBML), which is a derivative of Extensible Binary Meta Language (XML), or files encoded in accordance with the Moving Picture Experts Group (MPEG) standard; the program stream index may be provided in a Synchronized Multimedia Integration Language (SMIL) format; and client device **104** may download container files from CDN **114** over networks **106** using the HTTP protocol. In other embodiments, the container file formats may include OGG, flash video (FLV), Windows Media Video (WMV), or any other format.

(39) Exemplary, non-limiting, encoding bitrates for different levels, e.g., levels L1, L2, L3 may range from below 125 kilo-bits-per-second (kbps) up to 15,000 kbps, or even higher, depending on the type of encoded media (i.e., content). Video resolutions Res 1-Res 4 may be equal to or different from each other.

(40) The container files may support adaptive streaming of encoded video programs across an available spectrum bandwidth that is divided into multiple, i.e., n, levels. Video having a predetermined video resolution for each level may be encoded at a bitrate corresponding to the bandwidth associated with the given level. For example, in DivX® Plus Streaming, by Rovi Corporation, the starting bandwidth is 125 kbps and the ending bandwidth is 8400 kbps, and the number n of bandwidth levels is eleven (11). Each bandwidth level encodes a corresponding video stream, where the maximum encoded bitrate of the video stream (according to a hypothetical reference decoder model of the video coding standard H.264) is set equal to the bandwidth/bitrate of the given level. In DivX® Plus Streaming, the 11 levels are encoded according to 4 different video resolution levels, in the following way: mobile (2 levels), standard definition (4 levels), 720p (2 levels), and 1080p (3 levels).

(41) 2.1 Encoded Video Frame Structure

(42) FIG. 3A is an illustration of an example frame structure **300** of an encoded video block for container files from adaptive bitrate streams 1 and 2 of FIG. 2. Video encoding by encoder **110** includes capturing a number of successive picture frames, i.e., a GOP, at a predetermined video frame rate, and encoding each of the captured frames, in accordance with an encoding standard/technique, into a corresponding encoded video frame. Exemplary encoding standards include, but are not limited to, block encoding standards, such as H.264 and Moving Picture Experts Group (MPEG) standards. Collectively, the encoded video frames form an encoded video block, such as an encoded video block in one of container files CF. The process repeats to produce contiguous encoded video blocks.

(43) The encoding process may encode a video frame independent of, i.e., without reference to, any other video frames, such as preceding frames, to produce an encoded video frame referred to herein as a key frame. For example, the video frame may be intra-encoded, or intra-predicted. Such key frames are referred to as I-Frames in the H.264/MPEG standard set. Since the key frame was encoded independent of other encoded video frames, it may be decoded to recover the original video content therein independent of, i.e., without reference to, any other encoded video frames. In the context of streaming, the key frame may be downloaded from CDN **112** to client device **104**, decoded independent of other encoded frames, and the recovered (decoded) video played back, i.e., presented, on the client device.

(44) Alternatively, the encoding process may encode a video frame based on, or with reference to, other video frames, such as one or more previous frames, to produce an encoded video frame referred to herein as a non-key frame. For example, the video frame may be inter-encoded, i.e., inter-predicted, to produce the non-key frame. Such non-key frames include P-Frames and B-frames in the H.264/MPEG standard set. The non-key frame is decoded based on one or more other encoded video frames, e.g., key-frames, reference frames, etc. In the context of streaming, the non-key frame may be downloaded from CDN **112** to client device **104**, decoded based on other encoded frames, and the recovered video played back.

(45) With reference again to FIG. 3A, frame structure **300** of the encoded video block for container files in the adaptive bitrate streams includes, in a time-ordered sequence, a first set of successive non-key frames **304**, a key frame **306**, and a second set of successive non-key frames **308**. Accordingly, key frame **306** is interspersed among the encoded video frames of the encoded video block. The position of key frame **306** relative to the non-key frames in block **300** may vary, e.g., the position may be at the top, the middle, the bottom, or elsewhere in the block. Moreover, multiple key frames may be interspersed among the encoded video frames of the encoded video block, and separated from each other by multiple non-key frames.

(46) A key/non-key (K/NK) flag associated with each of the frames **304**, **306**, and **308** indicates whether the associated frame is a key-frame or a non-key frame. Each of the key and the non-key frames may include a predetermined number of bytes of encoded video.

(47) In an example in which the encoded video block represented by frame structure **300** encodes 2 seconds of video captured at a video frame rate of 30 frames per second (fps), the frame structure includes 60 encoded video frames, which may include N (i.e., one or more) interspersed key frames, and 60-N non-key frames. Typically, the number of non-key frames exceeds the number of key frames.

(48) FIG. 3B is an illustration of an example frame structure **320** of an encoded video block for container files from the trick play stream of FIG. 2. Trick play frame structure **320** includes, in a time-ordered sequence, key frames **322**. In other words, trick play frame structure **320** includes only key frames, i.e., key frames without non-key frames.

(49) In the example in which the encoded video block represented by frame structure **300** encodes 2 seconds of video captured at a video frame rate of 30 frames per second (fps), the encoded video block represented by frame structure **320** also encodes 2 seconds of video. However the video frame rate for structure **320** is reduced to 5 fps, which yields 10 encoded video frames (key frames) every 2 seconds.

(50) 3 Sequence Diagram

(51) FIG. 4 is a sequence diagram of example high-level interactions **400** between network services **102** and client device **104** used to initiate, i.e., startup, streaming, implement normal streaming and playback, and implement trick play features in on-demand, live, and real-time streaming embodiments. Interactions **400** progress in time from top-to-bottom in FIG. 4, and are now described in that order. It is assumed that prior to startup, encoder **110** is in the process of, or has finished, encoding video content into multiple adaptive bitrate streams and a corresponding trick play stream, and storing the resulting container files in CDN **112** for subsequent download to client device **104**.

(52) 3.1 Start-Up

(53) At **410**, a user of client device **104** selects content, such as a video program, to be streamed using the client device GUI.

(54) At **422**, client device **104** sends a “Start” message (also referred to as a “begin playback” message) to RTS **114** to start a streaming session. The Start message includes an identifier (ID) of the content to be streamed and a current time stamp. The ID identifies content from a content source that is to be streamed to client **104**, and may indicate, e.g., a channel, program name, and/or source originating the content to be streamed. The current time stamp (also referred to as “current time”) indicates a current time, such as a Universal Time Code (UTC). The UTC may be acquired from any available UTC time service, as would be appreciated by those of ordinary skill in the relevant arts.

(55) As mentioned above, it is assumed that at the time the Start message is issued, the content identified therein has already been encoded and is available for streaming, e.g., for video-on-demand streaming, or will begin to be encoded shortly after the time of the Start message, e.g., for live and real-time streaming. It is also assumed that RTS **114** has collected, or will be collecting, the information related to the encoded program from encoder **110** or CDN **115**, such as a program stream index, e.g., program stream index **204**, sufficient to identify the identified content in network services **102**.

(56) At **424**, in response to the Start message, RTS **114** sends an encoding profile message (referred to as a “Profile” message) to client **104**. The Profile message lists different encoding profiles used to encode the identified content, e.g., as available from the program stream index for the identified content. Each of the profiles specifies encoding parameters/settings, including, but not limited to: content type (e.g., audio, video, or subtitle); an encoding level corresponding to an encoding bitrate, resolution, and video frame rate (e.g., levels L1, L2, L3); and a container file type, e.g., a Multipurpose Internet Mail Extensions (MIME) type. The Profile message also indicates which encoding level among the multiple encoding levels e.g., encoding level L3, represents or corresponds to a trick play stream.

(57) In response to the Profile message, client device **104** selects an appropriate encoding level (e.g., an appropriate combination of an encoding bitrate and a resolution) among the levels indicated in the Profile message (not including the level indicating the trick play stream) for normal streaming and playback of the identified content. Client device **104** may determine the appropriate encoding level based on a communication bandwidth at the client device.

(58) 3.2 Normal Streaming and Playback

(59) After startup, normal streaming and playback begins, as follows.

(60) At **432**, after client device **104** has selected the encoding level, the client device sends a GetPlaylist message to RTS **114** to request a list of any new container files that have been uploaded since the client device last downloaded container files (if any) from CDN **112**. The GetPlaylist message includes selection criteria for uploaded container files, namely, a current time and the selected encoding level. The current time represents a time code associated with the last container file downloaded by client device **104** (if any) in the current streaming session.

(61) In response to the GetPlaylist message, RTS **114**: a. selects the uploaded container files, as identified to the RTS that meet the criteria specified in the GetPlaylist message. The selected, uploaded container files are those container files that have (i) a time code greater than the current time, and (ii) an encoding level that matches the level specified in the GetPlaylist message from the client device; b. generates a Playlist message identifying the selected container files; and c. at **433**, sends the Playlist message to client device **104**.

(62) For each of the selected container files, the Playlist message includes the following information: the type of content encoded in the container file (e.g., video, audio, or subtitle); an address (e.g., URL) of the container file in CDN **112** (e.g., a subset of the addresses A1 or A2); a time code, e.g., a start time and an end time, associated with the content block encoded in the container file; and a file size of the container file.

(63) At **434**, in response to the Playlist message, client device **104** downloads container files from addresses in CDN **112** based on, i.e., as identified in, the Playlist message.

(64) At **436**, client device **104** decodes all of the key frames and the non-key frames of the encoded content block from each of the downloaded container files to recover the original content therein, and then presents the recovered content, whether in audio, visual, or in other form, on client device **104**. The process of decoding the encoded content from the key and non-key frames and then presenting the recovered content on client device **104** is referred to as “normal playback” on the client device. In normal playback, the content recovered from successive downloaded container files is played back on client device **104** in a forward (play) direction, i.e., in an order of increasing time code. For example, with reference again to FIG. 2, the content is played back from container files CF in the time code order of 0 s-2 s, 2 s-4 s, 4 s-6 s, and so on. For normal playback, the decoded video frames are presented at a frame rate equal to the frame rate at which the video was original captured and encoded, e.g., at a rate of 30 fps.

(65) The normal streaming and playback sequence repeats. Therefore, in summary, in the streaming and playback sequence, client device **104** periodically requests and downloads Playlist messages, downloads container files indicated in the Playlist messages, and plays back the content from the downloaded container files in the forward direction.

(66) 3.3 Trick Play

(67) At any time during the normal streaming and playback sequence, the user may select a trick play (TP) feature through the GUI. Trick play features include, but are not limited to, rewind and fast forward, in which client device **104** rewinds and fast forwards through previously played back content.

(68) At **440**, assume the user selects the rewind trick play feature while client device **104** is performing the normal playback of content.

(69) At **442**, in response to the rewind request, client device **104** sends a GetPlaylist message to RTS **114** to solicit appropriate trick play video (container files) from network services **102**. Therefore, in this case, the GetPlaylist message may also be referred to as a “GetTrickPlayPlaylist” message. The GetPlaylist message sent at **442** includes the following trick play file selection criteria: a. a time (referred to as a “trick play time”) when the user selected the trick play feature; b. the encoding level that was indicated in the Profile message (at **424**) as corresponding to the trick play video (e.g., level 3 in the example of FIG. 2); and c. a trick play direction (depicted as “Dir” in FIG. 4) indicating rewind (RWD).

(70) At **444**, in response to the GetPlaylist message sent at **442**, RTS **114** generates and sends a trick play Playlist message to client device **104**. The trick play Playlist message identifies those container files from the trick play stream (e.g., the stream associated with encoding level L3 in the example of FIG. 2) that meet the selection criteria, namely, that are associated with (i) successive time code less than the trick play time because the trick play direction is RWD, and (ii) an encoding level that matches the specified level (e.g., encoding level L3). The Playlist message lists URLs of the appropriate trick play container files.

(71) At **446**, client device **104** downloads the trick play container files identified in the Playlist message from **444**. For example, client device **104** downloads the trick play container files from their corresponding URLs.

(72) At **448**, client device **104** plays back video from the downloaded trick play container files, i.e., the client device decodes the key frames from each of the trick play container files and then presents the decoded video in a rewind play direction, i.e., in an order of decreasing time codes beginning with the trick play time.

(73) The trick play sequence **442-448** repeats.

(74) During trick play, the video from the key frames may be played back at a reduced video frame rate relative to that used for normal playback. For example, the trick play playback video frame rate may be 5 fps, instead of 30 fps.

(75) Also, to implement a faster rewind, key frames may be skipped, e.g., every other key frame may be played back. In other words, only a subset of key frames in each of the downloaded trick play container files may be used in trick play playback.

(76) The above described trick play sequence results when the user selects RWD at **440**. Alternatively, the user may select fast forward (FFWD) at **440**. The trick play sequence that results when the user selects FFWD is similar to that for RWD, except that the GetPlaylist message at **442** indicates FFWD instead of RWD. In response to the FFWD indication in the GetPlaylist message, at **444**, RTS **114** returns a Playlist message identifying trick play files associated with successive time codes greater than (not less than) the trick play time. Then, at **448**, client device **104** plays back the

downloaded trick play files in the forward direction.

(77) 4 Profile and Playlist Messages

(78) 4.1 Profile Message

(79) FIG. 5 is an example Profile message **500**. In an embodiment, the Profile message format is in accordance with the World Wide Web Consortium (W3C) recommended Extensible Markup Language (XML) markup language, Synchronized Multimedia Integration Language (SMIL) 3.0 Tiny profile. This profile is well-suited to descriptions of web-based multimedia. However, other protocols may be used to format the Profile message.

(80) Profile message **500** includes a header **501** to specify the base profile as SMIL 3.0 (Tiny), and a body including video encoding (VE) profiles **502**, **504**, **505** and an audio encoding (AE) profile **506**. Profile message **500** corresponds to a requested program ID, such as encoded program **200** of FIG. 2, and includes information from the associated index, e.g., index **204**. Each of VE profiles **502**, **504**, **505** specifies the following encoding settings or parameters: a. a content type, e.g., video; b. an encoding level "Video ID" (e.g., level 1=L2, level 2=L2, level 3=L3) with its corresponding i. encoding bitrate (e.g., Rate1, Rate2, or Rate3, such as a bitrate=400000 bps, 600000 bps, or 150000 bps), and ii. video resolution (e.g., Res1, Res2, or Res3) in terms of, e.g., pixel width and height dimensions (e.g., 768×432); and c. MIME type.

(81) Similarly, AE profile **506** specifies: a. a content type, e.g., audio; b. an encoding bitrate/reserved bandwidth value (e.g., 192000); and c. a MIME type.

(82) The Profile message may also include a video frame rate at which each level was encoded.

(83) As mentioned above in connection with FIG. 4, Profile message **500** also includes a field **510** to indicate which of encoding profiles **502-505**, if any, represents a trick play stream. In the example of FIG. 5, the stream associated with level 3 (similar to FIG. 2) is indicated as the trick play stream.

(84) 4.2 Playlist Message

(85) FIG. 6 is an example Playlist message **600** generated in response to a GetPlaylist message selection criteria including a current time of 40 (seconds) and specifying a level 1 encoding level. Like the Profile message, the example Playlist message is formatted in accordance with SMIL 3.0.

(86) Playlist message **600** includes a header **601** to specify the base profile as 3.0, and a body that includes sequential records or elements **602-610**, each of which is defined as a seq element <seq>. In an embodiment, each seq element **602-610** corresponds to an uploaded container file. Using seq elements, RTS **114** is able to specify a sequence of real-time media streams for playback. A sequence tag is used with each element to indicate one of <video>, <audio> or <subtitle/text> encoded content for streaming. Elements **602-610** identify respective uploaded elements (e.g., container files) that meet the Playlist message criteria (i.e., encoding level 1 and a time code equal to or greater than 40). In the example of FIG. 6, elements **602-608** identify three container files containing successive or time-ordered two second blocks of encoded video. Element **610** identifies a container file containing a two second segment of encoded audio. Each of the Playlist message records **602-610** includes: a. a content type identifier (e.g., video or audio); b. a URL of the identified container file (e.g., src=http://10.180.14.232/1140.mkv). For example, the URLs correspond to container file addresses from the list of addresses A1 or A2 from FIG. 2; c. a time code in seconds (e.g., a start time and an end time, referred to as "ClipBegin" and "ClipEnd," respectively,) associated with the segment encoded in the identified container file. The example time codes for each of the container files are 40-42, 42-44, and 46-48; and d. a file size of the identified container file (e.g., 3200 kilobits).

5 Method Flowcharts

5.1 Network Side

(87) FIG. 7 is a flowchart of an example network-side method **700** of multimedia content streaming with trick play support based on trick play files, which may be implemented in network services **102**. Method **700** may be executed in accordance with sequence **400** of FIG. 4. The multimedia content includes video, and may also include audio and/or text (e.g., subtitles). Method **700** may be implemented in any of the contexts of on-demand, live, and real-time streaming.

(88) **715** includes encoding video into (i) multiple adaptive bitrate streams, and (ii) a corresponding trick play stream in accordance with corresponding distinct sets of encoder settings or levels, such as an encoding bitrate, a resolution, and a video frame rate. Each of the streams comprises container files of encoded video associated with successive time codes.

(89) **720** includes storing (i) the container files for each stream at corresponding addresses, such as network addresses, e.g., URLs, in a download server, e.g., in CDN **114**, and (ii) an index identifying the container files of each stream in RTS **114**.

(90) **725** includes receiving a playlist request (e.g., a GetPlaylist message) from a client device, e.g., over a communication network, for a selected one of the adaptive bitrate streams. The playlist request includes container file selection criteria, including a current time, an encoding level.

(91) **730** includes sending, to the client device over the communication network, a playlist (e.g., a Playlist message) identifying the stored files of the selected stream that meet the selection criteria, i.e., that are associated with time codes greater than the current time. The playlist may list URLs where the identified container files are stored and sizes of the files.

(92) **735** includes receiving, from the client device, a playlist request (e.g., another GetPlaylist message) for the trick play stream corresponding to the selected stream. The trick play playlist request includes a trick play time code, a trick play encoding level, and a trick play direction, e.g., fast forward or rewind.

(93) **740** includes sending, to the client device, a trick play playlist (e.g., another Playlist message) identifying the stored files (e.g., URLs of the stored files) of the trick play stream that are associated with successive time codes that are (i) less than the trick play time if the trick play direction is rewind, and (ii) greater than the trick play time if the trick play direction is fast forward.

(94) 5.2 Client Side

(95) FIG. 8 is a flowchart of an example client-side method **800** of multimedia content streaming with trick play support based on trick play files, which may be implemented in client device **104**. Method **800** is a client side method complementary to network side method **700**. Method **800** may be executed in accordance with sequence **400** of FIG. 4. The multimedia content includes video, and may also include audio and/or text (e.g., subtitles). Method **700** may be implemented in any of the contexts of on-demand, live, and real-time streaming.

(96) Together, operations **802-815** described below are considered precursor, or initialization, operations that lead to subsequent downloading of an adaptive bitrate stream.

(97) **802** includes requesting to stream a video program from network services over a communication network and, in response, receiving a Profile message over the communication network identifying multiple adaptive bitrate streams of encoded video and a trick play stream of encoded video that are stored in, and available for streaming from, network services. The streams may be identified according to their respective encoding levels (e.g., encoding bitrate, resolution, frame rate, etc.). Each of the streams comprises container files of the encoded video. The container files of each stream are associated with successive time codes.

(98) **805** includes selecting an adaptive bitrate stream from among the multiple adaptive bitrate streams. A client device may select an adaptive bitrate stream based on an available communication bandwidth.

(99) **810** includes sending, to the network services over the communication network, a playlist request (e.g., a GetPlaylist message) for (container) files from the selected stream. The playlist request includes file selection criteria that includes a current time and specifies an encoding level corresponding to, e.g., an encoding bitrate and a resolution, of the selected stream.

(100) **815** includes receiving, from the network services over the communication network, a playlist (e.g., a Playlist message) identifying the files from the selected stream that meet the file selection criteria, i.e., that are associated with successive time codes greater than the current time.

(101) **820** includes downloading, from the network services over the communication network, files of encoded video from the selected stream as

identified in the playlist, e.g., from URLs listed in the playlist.

(102) **825** includes playing back video from the downloaded files in an order of increasing time codes. This includes playing back video from both key and non-key frames at a normal video frame rate, such as 30 fps.

(103) **830** includes receiving a trick play feature request, such as a video rewind request, from a user of the client device. Next operations **835-850** are performed in response to the trick play request received at **830**.

(104) **835** includes sending, to the network services over the communication network, a trick play playlist request (e.g., a GetTrickPlayPlaylist message) for appropriate trick play files from the trick play stream corresponding to the selected stream. The request includes a trick play time (corresponding to a time when the user selected the trick play feature), a trick play encoding level as indicated in the Profile message received earlier by the client device at **802** (e.g., level L3), and a trick play direction (e.g., rewind or fast forward).

(105) **840** includes receiving, from the network services over the communication network, a trick play playlist (e.g., a Playlist message) identifying files from the trick play stream that meet the file selection criteria, i.e., that are associated with successive time codes (i) less than the trick play time if the direction is rewind, and (ii) greater than the trick play time if the direction is fast forward.

(106) **845** includes downloading the trick play files identified in the playlist from **840**, e.g., from URLs listed in the playlist.

(107) **850** includes playing back video from the downloaded files in either the rewind direction, i.e., in an order of decreasing time codes, or in the forward direction, as appropriate. This includes playing back video only from key frames at a trick play video frame rate, such as 5 fps, which is reduced relative to the normal frame rate.

(108) **6 Systems**

(109) FIG. **9A** is a block diagram of a computer system **900** configured to support/perform streaming and trick play features as described herein.

(110) Computer system **900** includes one or more computer instruction processing units and/or processor cores, illustrated here as processor **902**, to execute computer readable instructions, also referred to herein as computer program logic.

(111) Computer system **900** may include memory, cache, registers, and/or storage, illustrated here as memory **904**, which may include a non-transitory computer readable medium encoded with computer programs, illustrated here as computer program **906**.

(112) Memory **904** may include data **908** to be used by processor **902** in executing computer program **906**, and/or generated by processor **902** during execution of computer program **906**. Data **908** may include container files **908a** from adaptive bitrate streams and trick play streams, and message definitions **908b** for GetPlaylist, Playlist, and Profile messages, such as used in the methods described herein.

(113) Computer program **906** may include: Client application instructions **910** to cause processor **902** to perform client device functions as described herein. Instructions **910** include: GUI instructions **912** to implement a GUI through which a user may select to stream a program and select trick play features; streaming and playback instructions **914** to download, decode, and playback streamed video content; trick play instructions **916** to implement trick play features; and message protocol instructions **918** to implement client side message exchange protocols/sequences (sending and receiving of messages) as described in one or more examples above.

(114) Instructions **910-918** cause processor **902** to perform functions such as described in one or more examples above.

(115) FIG. **9B** is a block diagram of network/server-side application instructions **960** which may execute in a processing environment similar to that of computer system **900**, and which may be hosted in encoder **110**, RTS **114**, and/or CDN **112**, as appropriate.

(116) Network/server-side application instructions **960** cause a processor to perform network-side (network services) functions as described herein. Instructions **960** have access to adaptive bitrate streams, trick play streams, indexes identifying the streams, and message definitions as described in one or more example above. Instructions **960** include: encoder instructions **962** to encode multimedia content into adaptive bitrate streams and trick play streams, as described in one or more example above; and message protocol instructions **964**, including RTS instructions, to implement network side message exchange protocols/sequences (sending and receiving of messages) in support of adaptive bitrate streaming and trick play streaming, e.g., between RTS **114**, client device **104**, encoder **110**, and CDN **112**, as described in one or more examples above. For example, instructions **964** include instructions to create and send Profile and Playlist messages, and to respond to GetPlaylist messages.

(117) Methods and systems disclosed herein may be implemented with respect to one or more of a variety of systems including one or more consumer systems, such as described below with reference to FIGS. **10** and **11**. Methods and systems disclosed herein are not, however, limited to the examples of FIGS. **10** and **11**.

(118) FIG. **10** is a block diagram of an example computer system **1000** corresponding to any of network services **102**, including encoder **110**, CDN **112**, and RTS **114**. Computer system **1000**, which may be, e.g., a server, includes one or more processors **1005**, a memory **1010** in which instruction sets and databases for computer program applications are stored, a mass storage **1020** for storing, e.g., encoded programs, and an input/output (I/O) module **1015** through which components of computer system **1000** may communicate with communication network **106**.

(119) FIG. **11** is a block diagram of an example system **1100** representing, e.g., client device **104**, and may be implemented, and configured to operate, as described in one or more examples herein.

(120) System **1100** or portions thereof may be implemented within one or more integrated circuit dies, and may be implemented as a system-on-a-chip (SoC).

(121) System **1100** may include one or more processors **1104** to execute client-side application programs stored in memory **1105**.

(122) System **1100** may include a communication system **1106** to interface between processors **1104** and communication networks, such as networks **106**. Communication system **1106** may include a wired and/or wireless communication system.

(123) System **1100** may include a stream processor **1107** to process program (i.e., content) streams, received over communication channel **1108** and through communication system **1106**, for presentation at system **1100**. Stream processor **1107** includes a buffer **1107a** to buffer portions of received, streamed programs, and a decoder **1107b** to decode and decrypt the buffered programs in accordance with encoding and encryption standards, and using decryption keys. In an alternative embodiment, decoder **1107b** may be integrated with a display and graphics platform of system **1100**. Stream processor **1107** together with processors **1104** and memory **1105** represent a controller of system **1100**. This controller includes modules to perform the functions of one or more examples described herein, such as a streaming module to stream programs through communication system **1106**.

(124) System **1100** may include a user interface system **1110**.

(125) User interface system **1110** may include a monitor or display **1132** to display information from processor **1104**, such as a client-side GUI.

(126) User interface system **1110** may include a human interface device (HID) **1134** to provide user input to processor **1104**. HID **1134** may include, for example and without limitation, one or more of a key board, a cursor device, a touch-sensitive device, and/or a motion and/or image sensor. HID **1134** may include a physical device and/or a virtual device, such as a monitor-displayed or virtual keyboard.

(127) User interface system **1110** may include an audio system **1136** to receive and/or output audible sound.

(128) System **1100** may correspond to, for example, a computer system, a personal communication device, and/or a television set-top box.

(129) System **1100** may include a housing, and one or more of communication system **1106**, processors **1104**, memory **1105**, user interface system **1110**, or portions thereof may be positioned within the housing. The housing may include, without limitation, a rack-mountable housing, a desk-top housing, a lap-top housing, a notebook housing, a net-book housing, a set-top box housing, a portable housing, and/or other conventional electronic housing and/or future-developed housing. For example, communication system **1102** may be implemented to receive a digital television broadcast signal, and system **1100** may include a set-top box housing or a portable housing, such as a mobile telephone housing.

(130) Methods and systems disclosed herein may be implemented in circuitry and/or a machine, such as a computer system, and combinations thereof, including discrete and integrated circuitry, application specific integrated circuitry (ASIC), a processor and memory, and/or a computer-readable medium encoded with instructions executable by a processor, and may be implemented as part of a domain-specific integrated circuit

package, a system-on-a-chip (SOC), and/or a combination of integrated circuit packages. (131) Methods and systems are disclosed herein with the aid of functional building blocks illustrating functions, features, and relationships thereof. At least some of the boundaries of these functional building blocks have been arbitrarily defined herein for the convenience of the description. Alternate boundaries may be defined so long as the specified functions and relationships thereof are appropriately performed. While various embodiments are disclosed herein, it should be understood that they are presented as examples. The scope of the claims should not be limited by any of the example embodiments disclosed herein.

Claims

1. A playback device, comprising: a set of one or more processors; and a non-volatile storage containing an application for causing the set of one or more processors to perform normal playback of content by performing the steps of: receiving a playlist identifying a plurality of streams of content having different encoding bitrates for adaptive bitrate streaming, wherein the playlist comprises Uniform Resource Locators (URLs) of container files for the plurality of streams; performing adaptive bitrate streaming by switching between the plurality of streams in the received playlist, where performing adaptive bitrate streaming further comprises: downloading at least a portion of at least one file from one of the plurality of streams using at least one of the URLs from the playlist; and playing back video from the at least a portion of the at least one downloaded file; wherein the application is further for causing the set of one or more processors to perform trick play playback of content by performing the steps of: requesting a separate trick play playlist; receiving the trick play playlist comprising at least one URL for at least one trick play file from at least one trick play stream; downloading at least a portion of the at least one trick play file from the at least one trick play stream using the at least one URL from the trick play playlist, wherein the at least a portion of the at least one trick play file includes only key-frames in a time-ordered sequence; and playing back video from the key-frames from the at least a portion of the at least one downloaded trick play file.
2. The playback device of claim 1, wherein each of the plurality of streams and the trick play stream comprises segments of encoded video, wherein segments of encoded video from the plurality of streams comprise: encoded video frames, including non-key frames each encoded based on video from one or more previous video frames, and key-frames interspersed among the non-key frames, wherein each of the key-frames is encoded independently of previous video frames, and wherein segments of encoded video from the at least one trick play stream comprise encoded video frames, including only key-frames.
3. The playback device of claim 2, wherein: playing back video from the at least a portion of the at least one downloaded file includes playing back video from the non-key frames and the key-frames at a normal video frame rate; and playing back video from the at least a portion of the at least one downloaded trick play file includes playing back video from the key-frames at a trick play video frame rate.
4. The playback device of claim 3, wherein playing back video at the trick play video frame rate includes, selectively: playing back all of the video from each segment of encoded video from the at least a portion of the at least one downloaded trick play file to achieve a normal trick play playback rate; and playing back a subset of the video frames in each segment of encoded video from the at least a portion of the at least one downloaded trick play file to achieve an accelerated trick play playback rate.
5. The playback device of claim 1, wherein: the downloaded at least a portion of the at least one file from the one of the plurality of streams is associated with successive time codes; the playing back video from the at least a portion of the at least one downloaded file includes playing back the video in a forward direction of increasing time codes; the downloaded at least a portion of the at least one trick play file from the at least one trick play stream is associated with successive time codes; and the playing back video from the at least a portion of the at least one downloaded trick play file from the at least one trick play stream includes playing back the video in a rewind direction of decreasing time codes.
6. The playback device of claim 1, further comprising, prior to downloading of the at least a portion of the at least one file from one of the plurality of streams: sending a playlist request for segments of encoded video from the one of the plurality of streams associated with time codes greater than a current time specified in a playlist request; and receiving a playlist listing network addresses where the segments of encoded video from the one of the plurality of streams associated with successive time codes greater than the current time are stored, wherein the downloading includes downloading the segments of encoded video from the one of the plurality of streams from the network addresses, and the playing back video from the at least a portion of the at least one downloaded file from the one of the plurality of streams includes playing back video from the at least a portion of the at least one downloaded file in an order of increasing time codes.
7. The playback device of claim 1, wherein: the trick play playlist request is a request for a trick play playlist of segments of encoded video from the at least one trick play stream associated with time codes less than a trick play time when the trick play request was received; the trick play playlist lists network addresses of the segments of encoded video from the at least one trick play stream associated with time codes less than the trick play time; the downloading the at least a portion of the at least one trick play file from the at least one trick play stream include downloading segments of encoded video from the trick play stream from their network addresses; and the playing back video from the at least a portion of the at least one downloaded trick play file from the trick play stream includes playing back video in an order of decreasing time codes.
8. A method for playing back content on a playback device, comprising: receiving a playlist identifying a plurality of streams of content having different encoding bitrates for adaptive bitrate streaming, wherein the playlist comprises Uniform Resource Locators (URLs) of container files for the plurality of streams; performing adaptive bitrate streaming by switching between the plurality of streams in the received playlist, where performing adaptive bitrate streaming further comprises: downloading at least a portion of at least one file from one of the plurality of streams using at least one of the URLs from the playlist; and playing back video from the at least a portion of the at least one downloaded file; perform trick play playback of content by performing the steps of: requesting a separate trick play playlist; receiving the trick play playlist comprising at least one URL for at least one trick play file from at least one trick play stream; downloading at least a portion of the at least one trick play file from the at least one trick play stream using the at least one URL from the trick play playlist, wherein the at least a portion of the at least one trick play file includes only key-frames in a time-ordered sequence; and playing back video from the key-frames from the at least a portion of the at least one downloaded trick play file.
9. The method of claim 8, wherein each of the plurality of streams and the trick play stream comprises segments of encoded video, wherein segments of encoded video from the plurality of streams comprise: encoded video frames, including non-key frames each encoded based on video from one or more previous video frames, and key-frames interspersed among the non-key frames, wherein each of the key-frames is encoded independently of previous video frames, and wherein segments of encoded video from the at least one trick play stream comprise encoded video frames, including only key-frames.
10. The method of claim 9, wherein: playing back video from the at least a portion of the at least one downloaded file includes playing back video from the non-key frames and the key-frames at a normal video frame rate; and playing back video from the at least a portion of the at least one downloaded trick play file includes playing back video from the key-frames at a trick play video frame rate.
11. The method claim 10, wherein playing back video at the trick play video frame rate includes, selectively: playing back all of the video from each segment of encoded video from the at least a portion of the at least one downloaded trick play file to achieve a normal trick play playback rate; and playing back a subset of the video frames in each segment of encoded video from the at least a portion of the at least one downloaded trick play file to achieve an accelerated trick play playback rate.
12. The method of claim 8, wherein: the downloaded at least a portion of the at least one file from the one of the plurality of streams is associated with successive time codes; the playing back video from the at least a portion of the at least one downloaded file includes playing back the video in a forward direction of increasing time codes; the downloaded at least a portion of the at least one trick play file from the at least one trick play stream

is associated with successive time codes; and the playing back video from the at least a portion of the at least one downloaded trick play file from the at least one trick play stream includes playing back the video in a rewind direction of decreasing time codes.

13. The method of claim 8, further comprising, prior to downloading of the at least a portion of the at least one file from the one of the plurality of streams: sending a playlist request for segments of encoded video from the one of the plurality of streams associated with time codes greater than a current time specified in a playlist request; and receiving a playlist listing network addresses where the segments of encoded video from the one of the plurality of streams associated with successive time codes greater than the current time are stored, wherein the downloading includes downloading the segments of encoded video from the one of the plurality of streams from the network addresses, and the playing back video from the at least a portion of the at least one downloaded file from the one of the plurality of streams includes playing back video from the at least a portion of the at least one downloaded file in an order of increasing time codes.

14. The method of claim 8, wherein: the trick play playlist request is a request for a trick play playlist of segments of encoded video from the at least one trick play stream associated with time codes less than a trick play time when the trick play request was received; the trick play playlist lists network addresses of the segments of encoded video from the at least one trick play stream associated with time codes less than the trick play time; the downloading the at least a portion of the at least one trick play file from the at least one trick play stream include downloading segments of encoded video from the trick play stream from their network addresses; and the playing back video from the at least a portion of the at least one downloaded trick play file from the trick play stream includes playing back video in an order of decreasing time codes.
